

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

A motorcyclist rides off a horizontal platform with a velocity of 12 m s^{-1} . The platform is 10 m above the ground. Air resistance is negligible.

(a)
Calculate the time taken for the motorcyclist to reach the ground

Use vertical motion under constant acceleration.

Known values:

- $u = 0\text{ m s}^{-1}$ (no initial vertical velocity)
- $s = 10\text{ m}$
- $a = 9.8\text{ m s}^{-2}$

Use:

$$s = ut + \frac{1}{2}at^2 \Rightarrow 10 = 0 + \frac{1}{2} \times 9.8 \times t^2 \Rightarrow t^2 = \frac{20}{9.8} \Rightarrow t = \sqrt{2.04} = 1.43\text{ s}$$

(b)
Calculate the horizontal distance travelled by the motorcyclist

Use horizontal motion (no acceleration):

$$s = vt = 12 \times 1.43 = 17.2\text{ m}$$

(c)
Sketch the path of the motorcyclist and mark the velocity vector at two points

✔ The motorcyclist follows a **parabolic path** (projectile motion).
At the **start**, the velocity is **horizontal**: $\rightarrow$
As the motorcyclist falls, the vertical component increases due to gravity, so the velocity vector becomes **angled downward** near the end.

[ In your sketch: Show one arrow purely horizontal at launch, and one diagonally downward just before landing]

 Revision Tips

- Treat horizontal and vertical motions **separately**
- Vertical motion uses SUVAT equations (constant acceleration due to gravity)
- Horizontal motion has **constant velocity** (if no air resistance)
- Time to fall depends **only on vertical motion**
- Path is always a **symmetric parabola** if launched from level ground — not the case here since it's a horizontal launch from a height